

The Complete Fermat Sieve by Terminal Digit

A corrected and verified reference

Setup

For odd $N = p \cdot q$ with $1 \leq p < q$, write

$$R = \frac{p+q}{2}, \quad i = \frac{q-p}{2}, \quad \text{so that} \quad N = R^2 - i^2 = (R-i)(R+i).$$

Since N is odd, p, q have the same parity as each other only if that parity is odd (two odds multiply to odd; two evens multiply to even and are excluded). Concretely, R, i always have *opposite* parity, and which one is odd is determined by $N \bmod 4$:

$$N \equiv 1 \pmod{4} \iff R \text{ is odd, } i \text{ is even,} \quad (\text{Lane A})$$

$$N \equiv 3 \pmod{4} \iff R \text{ is even, } i \text{ is odd.} \quad (\text{Lane B})$$

In either lane, $2 \mid i$ or $2 \mid R$ respectively, so $4 \mid 2ab$ where $a = R, b = i$ — this divisibility is automatic and lane-independent.

This document corrects a specific error. A terminal-digit table indexed by $N \bmod 4$ and R 's own last digit (as in the original working note) is complete within each lane. But a table indexed by $N \bmod 10$ *alone* — with no reference to which lane it came from — must draw from *both* lanes simultaneously, because $N \bmod 10$ does not determine $N \bmod 4$: e.g. $N = 11$ and $N = 21$ are both $\equiv 1 \pmod{10}$, but $11 \equiv 3 \pmod{4}$ while $21 \equiv 1 \pmod{4}$. A version of this table that lists only 4 admissible $(R, i) \bmod 10$ classes per digit of N (rather than 8) has silently kept one lane and dropped the other. Every table below is verified by direct computation, not transcribed by hand.

1 Lane A: $N \equiv 1 \pmod{4}$ (R odd, i even)

Each row fixes $R \bmod 10$ and $i \bmod 10$ (respecting the required parities) and reports the resulting $N \bmod 10$, together with a genuine worked example $p \times q = N$.

$R \bmod 10$	$i \bmod 10$	$N \bmod 10$	Example (p, q)	$p \times q = N$
1	2	7	(9,13)	117
1	4	5	(7,15)	105
1	6	5	(5,17)	85
1	8	7	(3,19)	57
1	0	1	(1,21)	21
3	2	5	(1,5)	5
3	4	3	(9,17)	153
3	6	3	(7,19)	133
3	8	5	(5,21)	105
3	0	9	(3,23)	69
5	2	1	(3,7)	21
5	4	9	(1,9)	9
5	6	9	(9,21)	189
5	8	1	(7,23)	161
5	0	5	(5,25)	125
7	2	5	(5,9)	45
7	4	3	(3,11)	33
7	6	3	(1,13)	13
7	8	5	(9,25)	225
7	0	9	(7,27)	189
9	2	7	(7,11)	77
9	4	5	(5,13)	65
9	6	5	(3,15)	45
9	8	7	(1,17)	17
9	0	1	(9,29)	261

2 Lane B: $N \equiv 3 \pmod{4}$ (R even, i odd)

$R \bmod 10$	$i \bmod 10$	$N \bmod 10$	Example (p, q)	$p \times q = N$
2	1	3	(1,3)	3
2	3	5	(9,15)	135
2	5	9	(7,17)	119
2	7	5	(5,19)	95
2	9	3	(3,21)	63
4	1	5	(3,5)	15
4	3	7	(1,7)	7
4	5	1	(9,19)	171
4	7	7	(7,21)	147
4	9	5	(5,23)	115
6	1	5	(5,7)	35
6	3	7	(3,9)	27
6	5	1	(1,11)	11
6	7	7	(9,23)	207
6	9	5	(7,25)	175
8	1	3	(7,9)	63
8	3	5	(5,11)	55
8	5	9	(3,13)	39
8	7	5	(1,15)	15
8	9	3	(9,27)	243
0	1	9	(9,11)	99
0	3	1	(7,13)	91
0	5	5	(5,15)	75
0	7	1	(3,17)	51
0	9	9	(1,19)	19

3 The corrected, complete table indexed by $N \bmod 10$

This is the table that must combine both lanes. For each odd, non-multiple-of-5 last digit of N , there are **8** admissible $(R, i) \bmod 10$ classes, not 4 — four from each lane.

$N \equiv$	$(R, i) \bmod 10$	Lane	Example
1	(0,3)	B, $N \equiv 3(4)$	$7 \times 13 = 91$
	(0,7)	B, $N \equiv 3(4)$	$3 \times 17 = 51$
	(1,0)	A, $N \equiv 1(4)$	$1 \times 21 = 21$
	(4,5)	B, $N \equiv 3(4)$	$9 \times 19 = 171$
	(5,2)	A, $N \equiv 1(4)$	$3 \times 7 = 21$
	(5,8)	A, $N \equiv 1(4)$	$7 \times 23 = 161$
	(6,5)	B, $N \equiv 3(4)$	$1 \times 11 = 11$
	(9,0)	A, $N \equiv 1(4)$	$9 \times 29 = 261$
3	(2,1)	B, $N \equiv 3(4)$	$1 \times 3 = 3$
	(2,9)	B, $N \equiv 3(4)$	$3 \times 21 = 63$
	(3,4)	A, $N \equiv 1(4)$	$9 \times 17 = 153$
	(3,6)	A, $N \equiv 1(4)$	$7 \times 19 = 133$
	(7,4)	A, $N \equiv 1(4)$	$3 \times 11 = 33$
	(7,6)	A, $N \equiv 1(4)$	$1 \times 13 = 13$
	(8,1)	B, $N \equiv 3(4)$	$7 \times 9 = 63$
	(8,9)	B, $N \equiv 3(4)$	$9 \times 27 = 243$
7	(1,2)	A, $N \equiv 1(4)$	$9 \times 13 = 117$
	(1,8)	A, $N \equiv 1(4)$	$3 \times 19 = 57$
	(4,3)	B, $N \equiv 3(4)$	$1 \times 7 = 7$
	(4,7)	B, $N \equiv 3(4)$	$7 \times 21 = 147$
	(6,3)	B, $N \equiv 3(4)$	$3 \times 9 = 27$
	(6,7)	B, $N \equiv 3(4)$	$9 \times 23 = 207$
	(9,2)	A, $N \equiv 1(4)$	$7 \times 11 = 77$
	(9,8)	A, $N \equiv 1(4)$	$1 \times 17 = 17$
9	(0,1)	B, $N \equiv 3(4)$	$9 \times 11 = 99$
	(0,9)	B, $N \equiv 3(4)$	$1 \times 19 = 19$
	(2,5)	B, $N \equiv 3(4)$	$7 \times 17 = 119$
	(3,0)	A, $N \equiv 1(4)$	$3 \times 23 = 69$
	(5,4)	A, $N \equiv 1(4)$	$1 \times 9 = 9$
	(5,6)	A, $N \equiv 1(4)$	$9 \times 21 = 189$
	(7,0)	A, $N \equiv 1(4)$	$7 \times 27 = 189$
	(8,5)	B, $N \equiv 3(4)$	$3 \times 13 = 39$

The excluded case: $N \equiv 5 \pmod{10}$

$N \equiv 5 \pmod{10}$ means $5 \mid N$ — a structurally different case (it forces a factor of 5 into p or q , or into $R \pm i$), and is excluded from the tables above, matching the usual convention of building these sieves for sources coprime to the small primes handled separately elsewhere. For completeness: this residue has **18** admissible $(R, i) \pmod{10}$ classes (not 8), since the extra congruence $N \equiv 0 \pmod{5}$ removes one full constraint rather than adding one, so it is not simply "double" the other rows by lane alone.

Practical note

Any bound or sieve argument that filters admissible (R, i) (equivalently $(u, v) = (q_m - c_m, q_m + c_m)$) pairs by the last digit of N alone must use all 8 classes per digit (1,3,7, or 9), drawn from both parity lanes. Using only 4 undercounts the admissible set by half and any resulting bound derived from the 4-class version is not valid.